OOPs concepts in C++ language

2 Days Bootcamp

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Contents

- What is OOP ?
- Objects
- Class
- Difference between object and class
- Abstraction
- Encapsulation
- Inheritance
- Polymorphism

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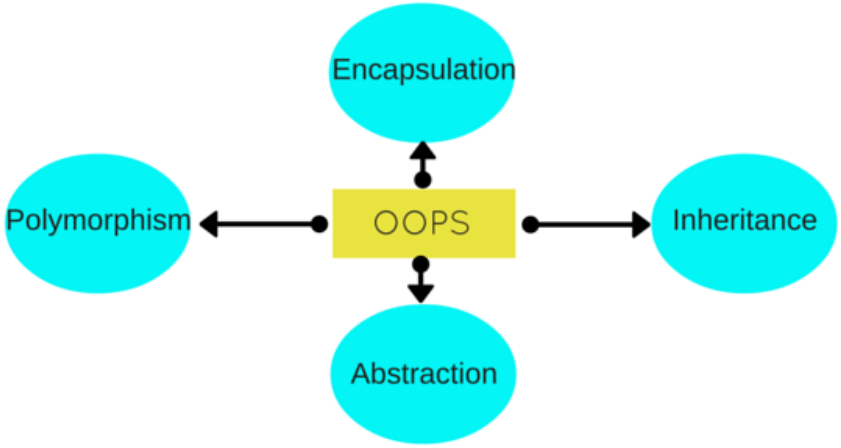
What is OOP?

- Object-oriented programming, or OOP, is an approach to problem solving where all computations are carried out using objects.
- programs are designed by making them out of objects that interact with one another.
- Significant object-oriented languages include
  - Java, C++, C#, Python, PHP, Ruby, Perl, Object Pascal, Objective-C, Dart, Swift, Scala, Common Lisp, and Smalltalk.



Objects

- Objects are the basic units of object-oriented programming.
- Objects contain
  - properties (data / fields / attributes)
  - functionality (methods)

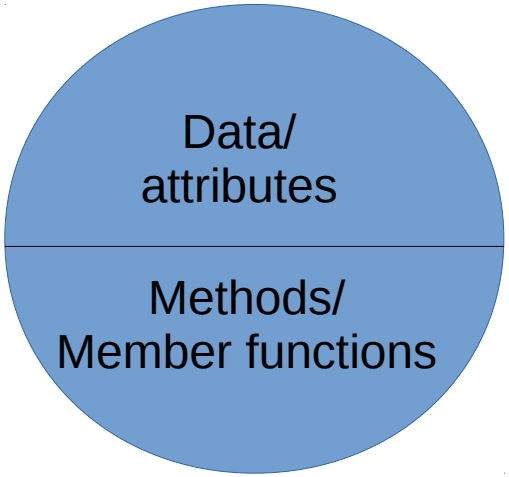
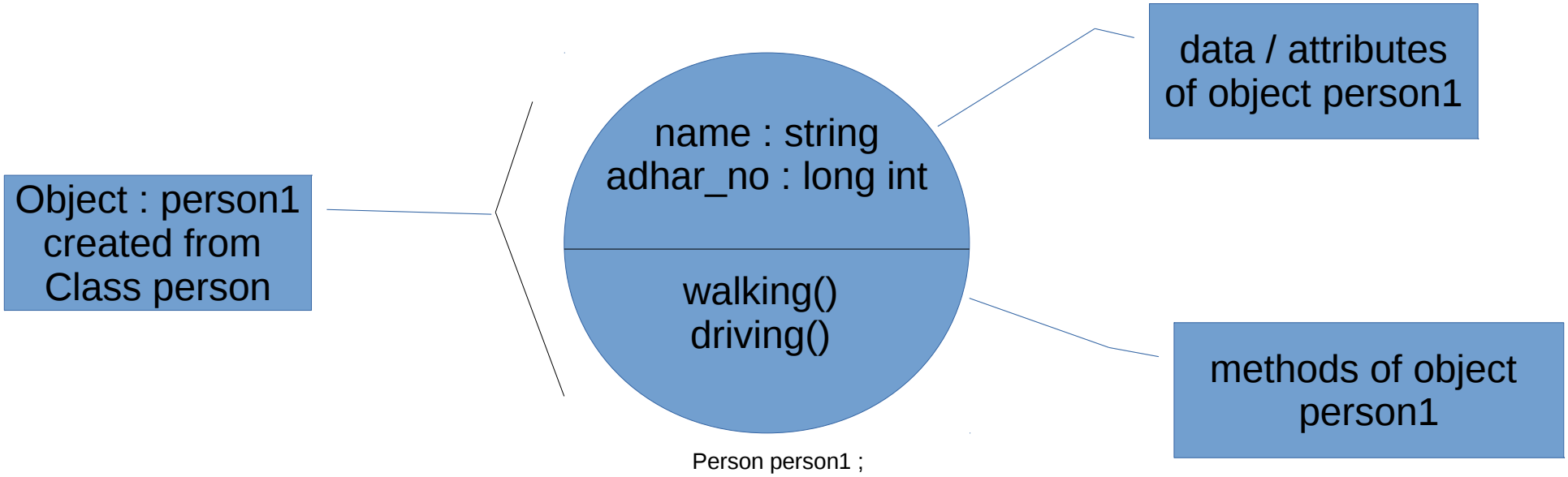


Fig : High level representation of an object

Objects continued..

- A simple example of an object would be a person.
  - Properties (data / attributes) – name, adhar no, etc.
  - Functionalities (methods) - walking, driving, etc.



- More examples of Objects:
  - Student, Car, Employee, Shape, Chair

Class

- It is a user defined data type.
- It holds its own data members and member functions, which can be accessed and used by creating an instance of that class (ie objects).
- **A class is like a blueprint for an object.**
- You can think of a class as a concept, and the object is the embodiment of that concept.
- **You need to have a class before you can create an object.**

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Diiference between class & Object

Object	Class
Object is an **instance** of a class.	Class is a **blueprint** or template from which objects are created.
Object is a **real world entity** such as pen, laptop, mobile, bed, keyboard, mouse, chair etc.	Class is a group of similar objects.
Object is a **physical** entity.	Class is a **logical** entity.
Object allocates memory when it is created.	Class doesn't allocated memory when it is created.

This difference will be more clear after reading next 2 slides..

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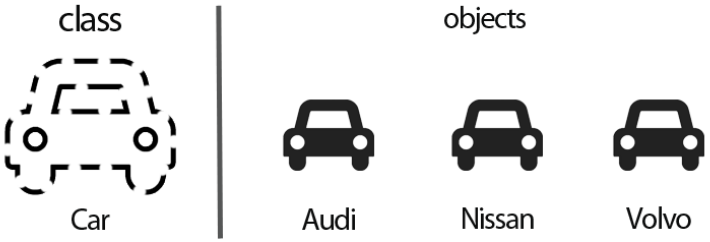
Example #1 :

- For example consider we have a **Class of Cars** under which Audi, Nissan and Volvo represents individual Objects.



- In this context each Car Object will have its own, Model,Year of Manufacture, Colour, Top Speed, Engine Power etc.,which form Properties of the Car class and the associated actions i.e., object functions like Start, Move, Stop form the Methods of Car Class.

- **No memory is allocated when a class is created. Memory is allocated only when an object is created, i.e., when an instance of a class is created.**

Example #2:

- An architect will have the blueprints for a house. those blueprints will be plans that explain exactly what properties the house will have and how they are all layed out.

- *However it is just the blueprint, you can't live in it.*

- Builders will look at the blueprints and use those blueprints to make a physical house. They can use the same blueprint to make as many houses as they want....each house will have the same layout and properties.

- Each house can accommodate it's own families...so one house might have the sameer live in it, one house might have the jyoti live in it.

- **The blueprint is the class...the house is the object. The people living in the house are data stored in the object's properties.**

Abstraction

- Abstraction means showing essential features and hiding non-essential features to the user.
- For Eg. gmail...
  - When you provide the user name and password and click on submit button
  - It will show Compose,Inbox,Outbox,Sentmails etc.
  - when you click on compose it will open
  - user doesn't know what are the actions performed internally.
- For Eg. Tv Remote..
  - Remote is a interface between user and tv.which has buttons like 0 to 10 ,on /of etc.
  - we dont know circuits inside remote.User does not need to know.

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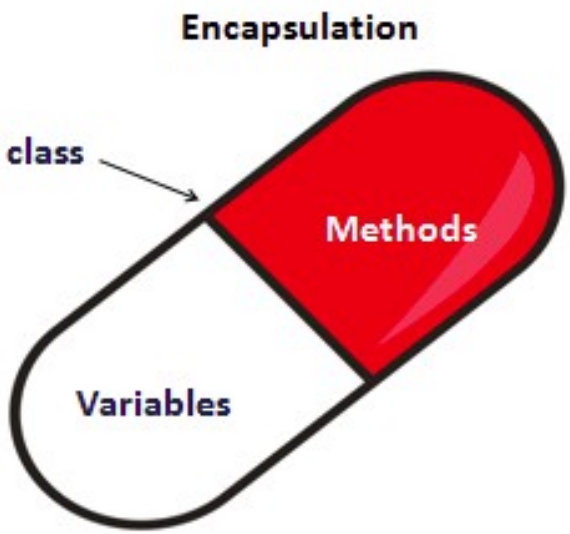
Encapsulation

- Encapsulation is binding the data (attributes) and behaviors (methods) together in a single unit (class) .
- For Example:
  - A car is having multiple parts like steering, wheels, engine etc. which binds together to form a single object that is car. So, Here multiple parts of cars encapsulates itself together to form a single object that is Car.
- In real time we are using Encapsulation for security purpose.
- Encapsulation = Abstraction + Data Hiding.



Inheritance

- In object-oriented programming, inheritance enables new objects to take on the properties of existing objects.
- A class that is used as the basis for inheritance is called a superclass or base class.
- A class that inherits from a superclass is called a subclass or derived class. The terms parent class and child class are also acceptable terms to use respectively.
- ***A child inherits visible properties and methods from its parent while adding additional properties and methods of its own.***

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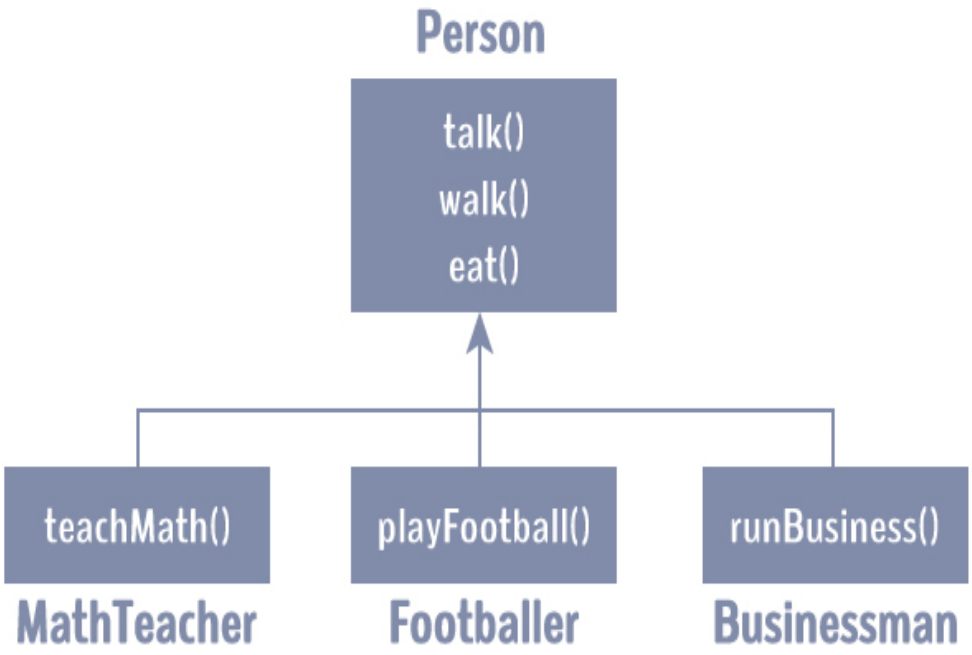
Inheritance example

- Person is a super / parent class.
- MathTeacher,Footballer, Businessman are derived / child classes.



Polymorphism

- polymorphism means ability to take more than one form.
- An operation may exhibit different behaviors in different instances.
- The behavior depends upon the types of data used in the operation.
- Ex-
  - We behave differently in front of elders, and friends.
    - A single person is behaving differently at different time.
  - A software engineer can perform different task at different instance of time depending on the task assigned to him .

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Polymorphism example

- If anybody says **CUT** to these people
  - The Surgeon
  - The Hair Stylist
  - The Actor
- What will happen?
  - The Surgeon would begin to make an incision.
  - The Hair Stylist would begin to cut someone's hair.
  - The Actor would abruptly stop acting out of the current scene, awaiting directorial guidance.
- So above representation shows What is polymorphism (same name , different behavior) in OOP.

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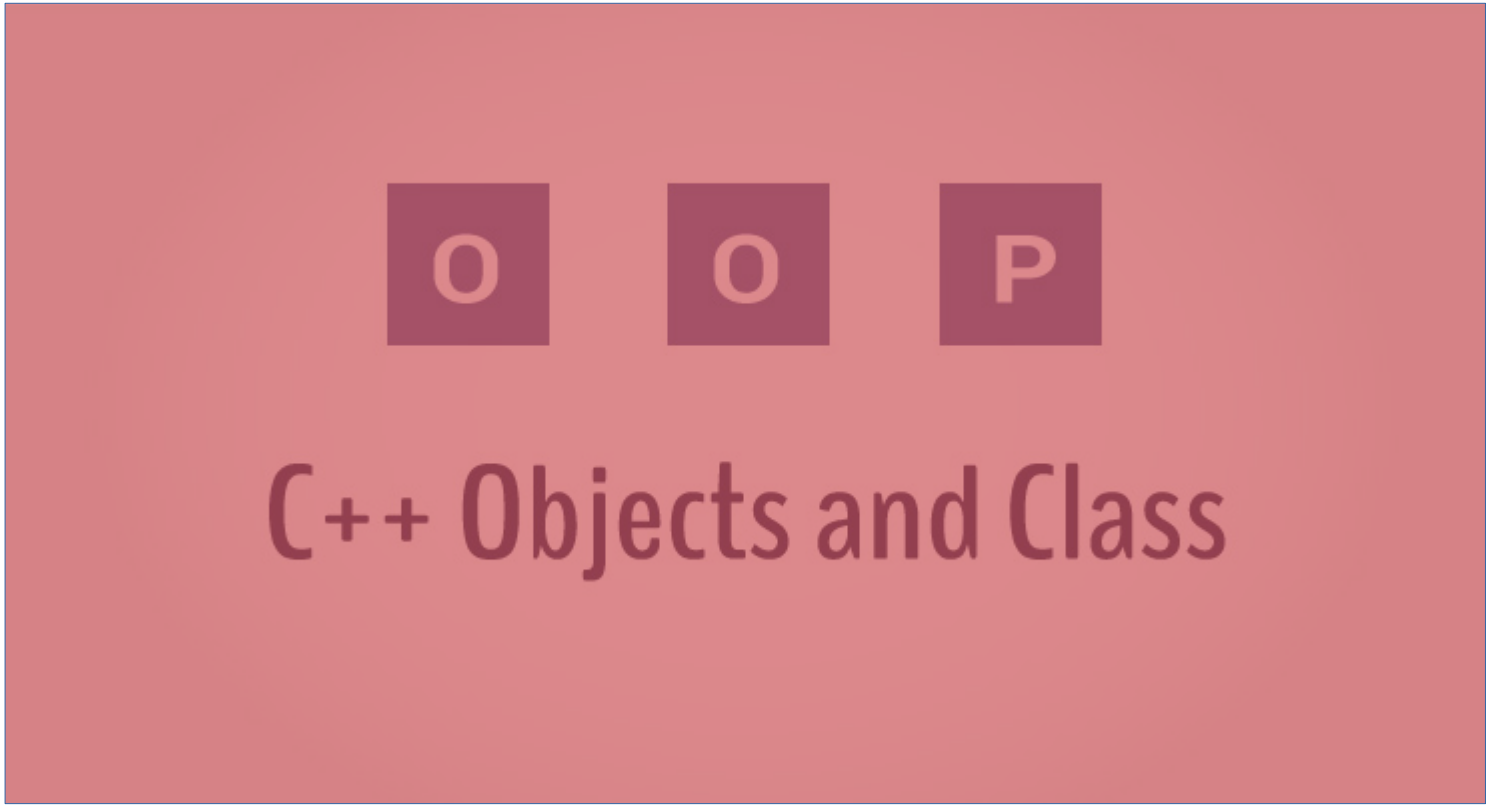
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C++ Objects and Class

C++ Class

- Before you create an object in C++, you need to define a class.

- **How to define a class in C++?**
  - A class is defined in C++ using **keyword class** followed by the **name of class.**
  - The body of class is defined inside the curly brackets and terminated by a semicolon at the end.

```
class className
{
        // some data
        // some functions
};
```


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Example #3: (person class)

```
#include<iostream>

#include<string>

using namespace std;

class Person {

    private:

        string name;

        long int adhar_no;

    public:

        void get_info(){

            cout<<"getting info..";

            cout<<"name : "<<name<<"\nadhar no : "<<adhar_no<<endl;

        }

        void set_info(string p_name,long int p_adhar_no){

            cout<<"setting info..\n";

            name = p_name;

            adhar_no = p_adhar_no;

        }

};
```

```
int main(){
Person person1 ;
person1.set_info("rushikesh phalke",12345678910);
person1.get_info();
}
```

Here, we defined a class named **Person** .

This class has
two data members: **name** and **adhar**_no.
two member functions: **set_info()** and **get_info()**

Keywords: private and public

- You may have noticed two keywords: private and public in the example#3.
- The private keyword makes data and functions private. Private data and functions can be accessed only from inside the same class.
- The public keyword makes data and functions public. Public data and functions can be accessed out of the class.
- Here, name and adhar_no are private members where as set_info() and get_info() are public members.
- If you try to access private data from outside of the class, compiler throws error. This feature in OOP is known as **data hiding.**

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C++ Objects

- When class is defined, only the specification for the object is defined; no memory or storage is allocated.

- To use the data and access functions defined in the class, you need to create objects.

- Syntax to Define Object in C++ :

```
className objectVariableName;
```

Example #3 revisited:

```
#include<iostream>

#include<string>

using namespace std;

class Person {

    private:

        string name;

        long int adhar_no;

    public:

        void get_info(){

            cout<<"getting info..";

            cout<<"name : "<<name<<"\nadhar no : "<<adhar_no<<endl;

        }

        void set_info(string p_name,long int p_adhar_no){

            cout<<"setting info..\n";

            name = p_name;

            adhar_no = p_adhar_no;

        }

};
```

Here, one object person1 of Person class is created.

```
int main(){

Person person1 ;

person1.set_info("rushikesh phalke",12345678910);
person1.get_info();

}
```


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How to access data member and member function in C++?

- We can access the data members and member functions by using a .(dot) operator.
- For example,

```
person1.get_info();
```

- This will call the get_info() function inside the Person class for object person1.
- Similarly, the data member can be accessed as:

```
person1.adhar_no = 12345678910;
```

- ***It is important to note that, the private members can be accessed only from inside the class (here adhar_no is private member)***.
- So, we can use person1.get_info(); from any function or class in the example#3. However, the code person1.adhar_no; should always be inside the class Person.

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Example #3 revisited:

```
#include<iostream>

#include<string>

using namespace std;

class Person {

    private:

        string name;

        long int adhar_no;

    public:

        void get_info(){

            cout<<"getting info..";

            cout<<"name : "<<name<<"\nadhar no : "<<adhar_no<<endl;

        }

        void set_info(string p_name,long int p_adhar_no){

            cout<<"setting info..\n";

            name = p_name;

            adhar_no = p_adhar_no;

        }

};
```

```
int main(){

Person person1 ;

person1.set_info("rushikesh phalke",12345678910);
person1.get_info();

person1.adhar_no=12345678910;

}
```

Accessing public members Outside class (WORKS)

Accessing private members outside class is not allowed.. (ERROR)